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PREPARATION METHODS OF POSITIVE ACTIVE MATERIALS, AND RECHARGEABLE LITHIUM BATTERIES

Abstract

Disclosed are a method of preparing a positive electrode active material, a positive electrode active material prepared according to the method, and a rechargeable lithium battery including the positive electrode active material. The method includes mixing a nickel-based precursor and a first lithium raw material, and performing a first heat treatment at about 500° C. to about 750° C. to prepare a first fired product, mixing the first fired product and a second lithium raw material, and performing a second heat treatment at about 650° C. to about 850° C. to prepare a positive electrode active material in the form of single particles including a lithium nickel-based composite oxide. A molar ratio (L.sup.1) of lithium in the first lithium raw material to a total metal of the nickel-based precursor is about 0.2 to about 0.9, a molar ratio (L.sup.2) of lithium in the second lithium raw material to a total metal excluding lithium in the first fired product is about 0.1 to about 0.8, about $0.9 \leq L.sup.1 + L.sup.2 \leq 1.1$.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority to Korean Patent Application No. 10-2024-0034116, filed in the Korean Intellectual Property Office on Mar. 11, 2024, and the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] Preparation methods of positive electrode active materials and rechargeable lithium batteries are disclosed.

2. Description of the Related Art

[0003] A portable information device such as, e.g., a cell phone, a laptop, smart phone, and the like, or an electric vehicle, typically uses a rechargeable lithium battery having high energy density and portability as a driving power source. Research is actively conducted to use a rechargeable lithium battery with high energy density as a driving power source or power storage power source for hybrid or electric vehicles.

[0004] Various positive electrode active materials can be used to manufacture rechargeable lithium batteries for the above applications. Among these positive electrode active materials, lithium nickel-based oxide, lithium nickel manganese cobalt composite oxide, lithium nickel cobalt aluminum composite oxide, and lithium cobalt oxide may constitute positive electrode active materials. Such nickel-based positive electrode active materials including nickel as a main transition metal are advantageous because the nickel-based positive electrode active materials may achieve high capacity, but there are limitations due to various challenges such as structural deterioration according to charges and discharges, a surface side reaction with an electrolyte, and deterioration due to particle cracks. For example, the nickel-based positive electrode active materials may present a challenge in that the higher nickel content, the higher the unreacted lithium content during the heat treatment, and in order to remove the unreacted lithium and control surface properties, a washing process may be required, but may result in losing lithium inside the positive electrode active materials during the washing, which may result in reducing capacity and deteriorating the surface properties. The surface properties may be restored by compensating the lost lithium, but the washing process and the compensation of the lithium raw material may significantly increase a material-processing cost. Accordingly, a method of manufacturing nickel-based positive electrode active materials implementing high performance, while improving surface properties may be advantageous.

SUMMARY

[0005] Some example embodiments provide a method of preparing a high-performance nickel-based positive electrode active material that realizes high capacity, desired or improved surface

properties, low residual lithium content, high pellet density, and long lifecycle characteristics, and is a novel preparation method with reduced processing costs and increased processability and productivity.

[0006] In some example embodiments, a method of preparing a positive electrode active material includes mixing a nickel-based precursor and a first lithium raw material and performing a first heat treatment at about 500° C. to about 750° C. to prepare a first fired product, mixing the first fired product and a second lithium raw material and performing a second heat treatment at about 650° C. to about 850° C. to prepare a positive electrode active material in a form of single particles including a lithium nickel-based composite oxide, wherein a molar ratio (L.sup.1) of lithium in the first lithium raw material to a total metal of the nickel-based precursor is about 0.2 to about 0.9, a molar ratio (L.sup.2) of lithium in the second lithium raw material to a total metal excluding lithium in the first fired product is about 0.1 to about 0.8, and about $0.9 \leq L.\text{sup.1} + L.\text{sup.2} < \text{about } 1.1$.

[0007] In some example embodiments, a positive electrode active material prepared according to the above method is provided.

[0008] In some example embodiments, a rechargeable lithium battery includes the positive electrode including a positive electrode active material, a negative electrode, and an electrolyte.

[0009] According to a method for preparing a positive electrode active material according to some example embodiments, processing costs can be reduced, processability and productivity can be increased, and high-capacity and long lifecycle characteristics can be realized while providing high-performance nickel-based positive electrode active material have desired or improved surface characteristics, low residual lithium content, and high pellet density.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIGS. 1 to 4 are schematic diagrams illustrating a rechargeable lithium battery according to some example embodiments.

[0011] FIG. 5 is a scanning electron microscope (SEM) image of the first fired product of Example.

[0012] FIG. 6 shows the results of Rietveld Refinement of X-ray Diffraction for the first fired product of Example.

[0013] FIGS. 7 and 8 are SEM images of the final positive electrode active material in the shape of single particles of Example.

DETAILED DESCRIPTION

[0014] Hereinafter, example embodiments will be described in detail so that those of ordinary skill in the art can readily implement the example embodiments. However, this disclosure may be embodied in many different forms and is not construed as limited to the example embodiments set forth herein.

[0015] The terminology used herein describes example embodiments only, and is not intended to limit the present disclosure. The singular expression includes the plural expression unless the context clearly dictates otherwise.

[0016] As used herein, “combination thereof” means a mixture, a laminate, a composite, a copolymer, an alloy, a blend, a reaction product, and the like of the constituents.

[0017] Herein, it should be understood that terms such as “comprises,” “includes,” or “have” are intended to designate the presence of an embodied feature, number, step, element, or a combination thereof, but it does not preclude the possibility of the presence or addition of one or more other features, number, step, element, or a combination thereof.

[0018] In the drawings, the thickness of layers, films, panels, regions, etc., are exaggerated for clarity and like reference numerals designate like elements throughout the specification. It will be

understood that when an element such as a layer, film, region, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

[0019] In addition, “layer” herein includes not only a shape formed on the whole surface when viewed from a plan view, but also a shape formed on a partial surface.

[0020] The average particle diameter may be measured by a method well known to those skilled in the art, for example, by a particle size analyzer, or by a transmission electron microscope image or a scanning electron microscope image. Alternatively, it is possible to obtain an average particle diameter value by measuring using a dynamic light scattering method, performing data analysis, counting the number of particles for each particle size range, and calculating from this. Unless otherwise defined, the average particle diameter may mean the diameter (D_{50}) of particles having a cumulative volume of 50 volume % in the particle size distribution. As used herein, when a definition is not otherwise provided, the average particle diameter means a diameter (D_{50}) of particles having a cumulative volume of 50 volume % in the particle size distribution that is obtained by measuring the size (diameter or major axis length) of about 20 particles at random in a scanning electron microscope image.

[0021] Herein, “or” is not to be construed as an exclusive meaning, for example, “A or B” is construed to include A, B, A+B, and the like.

[0022] “Metal” is interpreted as a concept including ordinary metals, transition metals and metalloids (semi-metals).

[0023] When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

Method of Preparing Positive Electrode Active Material

[0024] In some example embodiments, a method of preparing a positive electrode active material includes (i) mixing a nickel-based precursor and a first lithium raw material, and performing a first heat treatment on the nickel-based precursor and the first lithium raw material at about 500°C . to about 750°C . to prepare a first fired product, (ii) mixing the first fired product and a second lithium raw material, and performing a second heat treatment on the first fired product and the second lithium raw material at about 650°C . to about 850°C . to prepare a positive electrode active material in the form of single particles including a lithium nickel-based composite oxide. A molar ratio (L_{100}) of lithium in the first lithium raw material to a total metal of the nickel-based precursor is about 0.2 to about 0.9, a molar ratio (L_{200}) of lithium in the second lithium raw material to a total metal excluding lithium in the first fired product is about 0.1 to about 0.8, and about $0.9 \leq L_{100} + L_{200} \leq \text{about } 1.1$. Herein, the positive electrode active material can be expressed as a positive electrode active material for a rechargeable lithium battery or a nickel-based positive electrode active material.

[0025] The preparation method may be a no-washing method that does not include a water washing process. In other words, the method may be a novel method of effectively reducing unreacted lithium on the surface of the positive electrode active material, and strengthening the surface structure without going through the process of washing the positive electrode active material or intermediate products in the process of preparing the positive electrode active material. According to some example embodiments, the washing process can be omitted, thereby reducing processing costs, and the synthesis paradigm of the nickel-based positive electrode active material can be simplified. In addition, by dividing the lithium raw material into two or more portions, e.g., the first lithium raw material for the first fired product and the second lithium raw material for the final product, productivity can be improved by increasing a charging amount of raw materials into the reactor during firing. Furthermore, by dividing the lithium raw materials into the first lithium raw

material and the second lithium raw material, and then adding the first and second lithium raw materials after firing, the challenge of raw materials agglomerating can be effectively alleviated, thereby increasing process yield and improving productivity. According to the above method, it is readily advantageous to control the content relationship, such as by adjusting molar ratios of lithium in the second heat treatment process so that the final positive electrode active material includes an excessive amount of lithium, and adding coating raw materials, making it easier to modify the surface of the positive electrode active material and further strengthening the surface structure. In addition, generally, when preparing a nickel-based positive electrode active material in the form of single particles, a process of pulverizing the agglomerated particles to obtain a single particle shape is involved, but according to some example embodiments, the process can be simplified because a pulverizing process is not required, various process issues arising from reduced flowability of the powder during the pulverizing process can be resolved, and high-quality single particles of substantially uniform shape and size can be prepared. The positive electrode active material prepared by the above method can control the residual lithium on the surface to about 2000 ppm or less, the shape and size of the single particles can be substantially uniform, the single particle surface is substantially clean and the Brunauer, Emmett and Teller (BET) specific surface area is small, which is advantageous for improving the electrode density and electrolyte impregnation speed.

[0026] In examples, the single particles may exist alone without a grain boundary within the particle, is composed of one particle, and may be or include a monolith structure, a one body structure, or a non-agglomerated particle, in which particles are not agglomerated with each other but exist as an independent phase in terms of morphology, and may be expressed as one body particle, single grain, for example, as a single crystal. The single particles may be alone, or single particles may be agglomerated together. For example, 2 to 10 single particles may be agglomerated and in contact with each other.

[0027] The method for preparing a positive electrode active material according to some example embodiments includes adding the lithium raw material two or more times, for example adding a first lithium raw material in process (i) and adding a second lithium raw material in process (ii). Herein, a molar ratio ($L_{sup.1}$) of lithium in the first lithium raw material to a total metal of the nickel-based precursor may be about 0.2 to about 0.9, for example about 0.3 to about 0.8, about 0.4 to about 0.7, about 0.5 to about 0.7, or about 0.5 to about 0.6. In addition, the molar ratio ($L_{sup.2}$) of lithium in the second lithium raw material to a total metal excluding lithium in the first fired product may be about 0.1 to about 0.8, for example, about 0.2 to about 0.7, about 0.3 to about 0.6, about 0.4 to about 0.6, or about 0.4 to about 0.5. A sum of $L_{sup.1} + L_{sup.2}$ may indicate a molar ratio of lithium to a total metal excluding lithium in the final positive electrode active material, and the range may be about $0.9 \leq L_{sup.1} + L_{sup.2} \leq \text{about } 1.1$, about $0.95 \leq L_{sup.1} + L_{sup.2} \leq \text{about } 1.05$, about $0.9 \leq L_{sup.1} + L_{sup.2} \leq \text{about } 1.0$, about $0.9 \leq L_{sup.1} + L_{sup.2} \leq \text{about } 0.995$, or about $1.0 \leq L_{sup.1} + L_{sup.2} \leq \text{about } 1.1$. In this case, $L_{sup.1} > L_{sup.2}$ can be satisfied, and as another example, it can be designed to satisfy $L_{sup.1} \leq L_{sup.2}$. By appropriately adjusting $L_{sup.1}$ and $L_{sup.2}$ and appropriately controlling the heat treatment temperature, a nickel-based positive electrode active material that realizes high capacity and long lifecycle can be effectively obtained.

[0028] The nickel-based precursor may include, for example, a nickel-based hydroxide, a nickel-based oxide, or a combination thereof. For example, the nickel-based hydroxide can be synthesized through a coprecipitation reaction, and the nickel-based oxide can be synthesized by a method such as, e.g., calcining the nickel-based hydroxide in an oxidizing atmosphere.

[0029] For example, the nickel-based precursor may be represented by Chemical Formula 11 or Chemical Formula 12.

$Ni_{x11}M_{y11}M_{z11}(OH)_{sub.2}$ Chemical Formula 11

[0030] In Chemical Formula 11, $0.7 \leq x_{11} < 1$, $0 < y_{11} \leq 0.3$, $0 \leq z_{11} \leq 0.3$, $0.9 \leq x_{11} + y_{11} + z_{11} \leq 1.1$, and

M.sup.11 and M.sup.12 each independently are or include one or more of Al, B, Ba, Ca, Ce, Co, Cr, Cu, Fe, Mg, Mn, Mo, Nb, Si, Sn, Sr, Ti, V, W, Y, Zn, and Zr.

[0031] In Chemical Formula 11, $0.7 \leq x_{11} \leq 0.99$, $0.01 < y_{11} \leq 0.3$, $0 \leq z_{11} \leq 0.3$, $0.8 \leq x_{11} \leq 0.99$, $0.01 < y_{11} \leq 0.2$, $0 \leq z_{11} \leq 0.2$, $0.85 \leq x_{11} < 1$, $0 < y_{11} \leq 0.15$, $0 \leq z_{11} \leq 0.15$, or $0.9 \leq x_{11} < 1$, $0 < y_{11} \leq 0.1$, $0 \leq z_{11} \leq 0.1$.

Ni.sub.x12M.sup.13.sub.y12M.sup.14.sub.z12O.sub.2-b12X.sub.b12 Chemical Formula 12

[0032] In Chemical Formula 12, $0.7 \leq x_{12} < 1$, $0 < y_{12} \leq 0.3$, $0 \leq z_{12} \leq 0.3$, $0.9 \leq x_{12} + y_{12} + z_{12} \leq 1.1$, $0 \leq b_{12} \leq 0.1$, M.sup.13 and M.sup.14 each independently are or include at least one of Al, B, Ba, Ca, Ce, Co, Cr, Cu, Fe, Mg, Mn, Mo, Nb, Si, Sn, Sr, Ti, V, W, Y, Zn, and Zr, and X is or includes at least one of F, P, and S.

[0033] In Chemical Formula 12, $0.7 \leq x_{12} \leq 0.99$, $0.01 < y_{12} \leq 0.3$, $0 \leq z_{12} \leq 0.3$, $0.8 \leq x_{12} \leq 0.99$, $0.01 < y_{12} \leq 0.2$, $0 \leq z_{12} \leq 0.2$, $0.85 \leq x_{12} < 1$, $0 < y_{12} \leq 0.15$, $0 \leq z_{12} \leq 0.15$, or $0.9 \leq x_{12} < 1$, $0 < y_{12} \leq 0.1$, $0 \leq z_{12} \leq 0.1$.

[0034] According to some example embodiments, the nickel-based precursor has the same or similar particle size as the final single particle, and may be or include a precursor in the form of secondary particles in which a plurality of primary particles are agglomerated. For example, at least a portion or all of the primary particles may be needle-shaped. That is, according to some example embodiments, the method may include obtaining a positive electrode active material in the form of a single particle through an appropriate sintering process while using a precursor in the form of secondary particles having the same or similar particle size as the particle size of the single particle. This is different from the conventional method, which uses a precursor in the form of single particles, and requires a pulverizing process due to agglomeration occurring during the firing process. According to some example embodiments, a pulverizing process is not required, and a single particle shape with improved agglomeration can be synthesized by minimizing an amount of lithium used in each stage of synthesis from the precursor to the fired product and the final product, so that synthesis can proceed under mild conditions, and single particles with uniform size and shape and clean surfaces can be obtained.

[0035] The average particle diameter (D.sub.50) of the nickel-based precursor may be about 1 μm to about 6 μm , for example about 2 μm to about 6 μm , or about 2 μm to about 5 μm . When the precursor has a size within the ranges, as processability is improved, capacity may be increased. For example, when the precursor has too small a size, the precursor is difficult to disperse, which may deteriorate the processability, but when the size is too large, the capacity may be reduced. Herein, the average particle diameter (D.sub.50) indicates a size of particles having a cumulative volume of 50 volume % in the particle size distribution that is obtained by measuring the size of about 20 particles at random from a scanning electron microscope (SEM) image.

[0036] For example, a difference between the average particle diameter (D.sub.50) of the nickel-based precursor and the average particle diameter (D.sub.50) of the final positive electrode active material in the form of single particles may be less than or equal to about 1 μm , or less than or equal to about 0.5 μm , for example, about 0 μm to about 1 μm , or about 0.01 μm to about 0.5 μm .

[0037] The nickel-based precursor may further include zirconium in addition to the nickel. In other words, the nickel-based precursor may be or include hydroxide or oxide including nickel and zirconium. In other words, the nickel-based precursor may be or include a zirconium-doped nickel-based precursor. Herein, the zirconium may be a type of dopant and, and may also play the role of a grain growth accelerator during the heat treatment. When the zirconium-doped nickel-based precursor is used, the zirconium-doped nickel-based precursor may promote grain growth, making it possible to effectively synthesize single particles at a lower temperature than the conventional synthesis methods of single particles, and thereby reduce or suppress agglomeration of particles, thus improving productivity. Existing alkaline grain growth accelerators or fluxes had the problem of remaining after firing and acting as resistance within the positive electrode, thereby reducing the

lifecycle thereof. However, when a zirconium-doped nickel-based precursor is used as a raw material, impurities may not remain on the surface of the positive electrode active material particles, and lifespan characteristics can thus be improved. In some example embodiments, the zirconium-doped nickel-based precursor may be used to prepare a high nickel-based positive electrode active material in the form of single particles at a relatively low fire temperature in a simple method without adding the alkali-based grain growth accelerator or flux, and the like, improving productivity and economy.

[0038] The first lithium raw material may be or include, for example, at least one of lithium hydroxide, lithium carbonate, lithium sulfate, lithium nitrate, or a combination thereof, and for example, the first lithium raw material may be or include anhydrous lithium hydroxide. The types of the first lithium raw material and the second lithium raw material may be the same or different, for example, the first lithium raw material and the second lithium raw material may be the same as each other.

[0039] When anhydrous lithium hydroxide is the lithium raw materials, a charged amount of the raw materials may be increased in the synthesis process of a positive electrode active material, improving hourly production but decreasing unnecessary gas and moisture generated during the heat treatment, and as a result improving processability and enhancing quality of the positive electrode active material. In addition, there may be no input of unnecessary heavy substances such as, e.g., H_2O and the like, increasing a heat treatment yield and improving productivity.

[0040] For example, the anhydrous lithium hydroxide (LiOH) may be prepared by drying hydrated lithium hydroxide ($\text{LiOH} \cdot \text{H}_2\text{O}$) with an average particle diameter (D_{50}) of about 400 μm to about 600 μm , and pulverizing the dried hydrated lithium hydroxide into an average particle diameter (D_{50}) of about 3 μm to about 30 μm . The anhydrous lithium hydroxide may not be pulverized before the drying, and may be pulverized once for about 1 minute after the drying. The drying may be, for example, performed under a vacuum condition within a temperature range of about 50° C. to about 200° C. for a period of about 0.5 to about 20 hours. The average particle diameter (D_{50}) of the hydrated lithium hydroxide, which is a starting material, may be, for example, about 450 μm to about 550 μm or about 480 μm to about 500 μm , but the obtained anhydrous lithium hydroxide may have an average particle diameter (D_{50}) of about 3 μm to about 25 μm or about 5 μm to about 20 μm .

[0041] The method of manufacturing the anhydrous lithium hydroxide may make it possible to readily obtain anhydrous lithium salt, maintain an optimal process condition, and reduce a conversion rate to Li_2CO_3 to about 5% or less, obtaining high-purity anhydrous lithium hydroxide. After pulverizing the anhydrous lithium hydroxide, because fluidity of the powder rapidly decreases, it may be generally difficult to perform an additional process. For example, when drying is performed after the pulverizing, the fine particles may be entangled with each other and tightly agglomerated by heat generated during the drying, which may require a re-pulverizing process, but the agglomerated particles may be more difficult to grind due to their high agglomeration strength in the re-pulverizing process. In addition, as the number of processes increases, the conversion rate to Li_2CO_3 also increases due to an increase in the specific surface area, which may result in failing to obtain high-quality anhydrous lithium hydroxide. However, the method of manufacturing anhydrous lithium hydroxide according to some example embodiments, in which the anhydrous lithium hydroxide is once pulverized into a specific size under a predetermined condition after the drying, is a simple process of obtaining high-quality anhydrous lithium hydroxide, and additional processes may be readily added thereto.

[0042] The first heat treatment may be performed within a temperature range of about 500° C. to about 750° C. When the first heat treatment is performed within the temperature range, a first fired product may have appropriately low crystallinity and a hexagonal crystal structure and a cubic crystal structure to secure a large number of lithium channels, which may increase reactivity with the second lithium raw material, and thus effectively promote particle growth. A positive electrode

active material prepared by applying the temperature conditions may exhibit desired or improved surface characteristics, a single particle shape with a uniform shape and size, a low residual lithium content, and high pellet density. When the first heat treatment temperature is greater than about 750° C., the first fired product may have a different crystal structure and different properties from the crystal structure and the properties of some example embodiments, and the fired product has relatively high crystallinity and less reactivity with lithium during the secondary firing, which may result in failing to obtain the desired high-quality single particles.

[0043] The first heat treatment may be performed, for example, at a temperature range of about 550° C. to about 750° C., about 600° C. to about 750° C., about 650° C. to about 750° C., or about 700° C. to about 750° C. For example, the first heat treatment temperature may be lower than the second heat treatment temperature described below. The first heat treatment may be performed for a period of about 6 to about 12 hours, or about 8 to about 12 hours, and may be performed in an oxidizing atmosphere. When the first heat treatment proceeds within any of the above temperature ranges for the above periods of time, a positive electrode active material in the form of single particles having desired or improved surface characteristics, high pellet density due to a uniform shape and size, and a low residual lithium content, may be effectively prepared.

[0044] In some example embodiments, when the nickel-based first lithium raw material is mixed, a zirconium raw material may be mixed together with the nickel-based first lithium raw material. Zirconium of the zirconium raw material may be a type of dopant, and may play a role of a grain growth accelerator. When the zirconium raw material is added, the zirconium raw material may promote grain growth, making it possible to effectively synthesize single particles at a lower temperature than the conventional synthesis methods of single particles, which may reduce or suppress agglomeration of particles, thereby, improving productivity. Conventional alkali-based grain growth accelerators or fluxes present a challenge of remaining after the firing and acting as resistance inside a positive electrode, and thus deteriorating a lifecycle, but the zirconium raw material is a dopant of the positive electrode active material and does not remain on the positive electrode active material particle surface, thereby improving lifecycle characteristics. In some example embodiments, the zirconium raw material is configured to effectively prepare a high nickel-based positive electrode active material in the form of single particles without adding the alkali-based grain growth accelerators or fluxes, etc. at a relatively low fire temperature in a simple method, which may improve productivity and economy.

[0045] A zirconium content of the zirconium raw material may be about 0.01 mol % to about 1 mol %, for example about 0.01 mol % to about 0.9 mol %, about 0.05 mol % to about 0.7 mol %, about 0.1 mol % to about 0.5 mol %, or about 0.1 mol % to about 0.3 mol % based on 100 mol % of a total metal of the nickel-based precursor and the zirconium of the zirconium raw material. In addition, a zirconium content of the zirconium raw material may be about 0.01 wt % to about 1 wt %, for example about 0.01 wt % to about 0.9 wt %, about 0.05 wt % to about 0.7 wt %, about 0.1 wt % to about 0.5 wt %, or about 0.1 wt % to about 0.3 wt % based on 100 wt % of a total metal of the nickel-based precursor and the zirconium of the zirconium raw material. When the zirconium content satisfies any of the above ranges, single particles of uniform size of about 1 μm to about 6 μm can be effectively prepared at low temperatures.

[0046] The zirconium raw material is a compound including zirconium and may be or include, for example, zirconium oxide (ZrO₂).

[0047] The preparing method of a positive electrode active material according to some example embodiments may further include a pulverizing process after the first heat treatment. Herein, the pulverizing indicates, for example, grinding powder with a size of 1 mm or less into microscale particles with a size of less than or equal to about 10 μm, and may be distinguished from crushing, which refers to breaking up a solid of greater than or equal to about 10 mm in size into powder of less than or equal to about 1 mm. The pulverizing may be carried out, for example, through a mild jet-mill or an air classifier mill (ACM).

[0048] The first fired product obtained through the first heat treatment may include lithium nickel-based composite oxide in the form of secondary particles formed by agglomerating a plurality of primary particles. The average particle diameter ($D_{sub.50}$) of the secondary particles that constitute the first fired product may have the same or similar size as the single particles that constitute the final positive electrode active material, for example, about 1 μm to about 6 μm , or about 2 μm to about 5 μm . For example, a difference between the average particle diameter ($D_{sub.50}$) of the first fired product and the average particle diameter ($D_{sub.50}$) of the final positive electrode active material in the form of single particles may be less than or equal to about 1 μm , or may be less than or equal to about 0.5 μm , for example, about 0 μm to about 1 μm , or about 0.01 μm to about 0.5 μm .

[0049] For example, the lithium nickel-based composite oxide of the first fired product may have a hexagonal crystal structure and a cubic crystal structure. For example, the first fired product may include about 60% to about 95%, about 70% to about 95%, or about 75% to about 90% of the hexagonal crystal structure, and about 5% to about 40%, about 5% to about 30%, or about 10% to about 25% of the cubic crystal structure. In this case, the first sintered product has high reactivity with the second lithium raw material and is advantageous for growing into single particles of substantially uniform size in the range of about 1 μm to about 6 μm , and can effectively reduce residual lithium on the surface of the final positive electrode active material. The crystal structure of the first fired product can be analyzed, for example, by X-ray diffraction Rietveld refining method.

[0050] The second lithium raw material may be or include, for example, at least one of lithium hydroxide, lithium carbonate, lithium sulfate, lithium nitrate, or a combination thereof, for example, anhydrous lithium hydroxide. The types of the first lithium raw material and the second lithium raw material may be the same or different, for example, the first lithium raw material and the second lithium raw material may be the same as each other. In the first fired product, the molar ratio ($L_{sup.2}$) of lithium in the second lithium raw material to a total metal excluding lithium may be in the range of about 0.1 to about 0.8 as described above, and for example about 0.2 to about 0.7, about 0.3 to about 0.6, about 0.4 to about 0.6, or about 0.4 to about 0.5. $L_{sup.2}$ may be appropriately adjusted to prepare a high-performance positive electrode active material in the form of single particles that has high lithium usable capacity, low surface residual lithium content, and is easier to crush.

[0051] The second heat treatment may be performed within a temperature range of about 650° C. to about 850° C., for example, about 700° C. to about 830° C., or about 750° C. to about 810° C. In addition, the second heat treatment may proceed for a period of about hours to about 16 hours, or for about 12 hours to about 16 hours, under an oxidizing atmosphere. When the second heat treatment proceeds under the above conditions, single particles with a uniform size in the range of about 1 μm to about 6 μm may be prepared, contributing to obtaining a positive electrode active material having high lithium usable capacity and a low residual lithium content on the surface.

[0052] According to some example embodiments, when the first fired product is mixed with the second lithium raw material, a coating raw material may also be mixed together with the first fired product and the second lithium raw material to proceed with coating the positive electrode active material particle surface in a simple method. The coating raw material may include, for example, one or more of Al, B, Co, Mg, V, Zn, and Zr. A content of the coating elements may vary based on types of elements, for example, about 0.1 wt % to about 5 wt % or about 0.5 wt % to about 3 wt % based on 100 wt % of total metal excluding lithium in the positive electrode active material. For example, the method of preparing a positive electrode active material may include mixing the first fired product with the second lithium raw material and the cobalt raw material together, and then performing the secondary heat treatment on the mixture to provide a positive electrode active material with improved lifecycle characteristics. Herein, a content of cobalt as the coating element may be about 0.1 wt % to about 3 wt %, or about 0.5 wt % to about 2 wt % based on 100 wt % of

the total metal excluding lithium in the positive electrode active material.

[0053] The method for preparing a positive electrode active material according to some example embodiments may further include a pulverizing process after the second heat treatment. The pulverizing may be carried out, for example, through a mild jet-mill or airflow grinder. Grinding productivity can be increased by applying mild pulverizing.

[0054] The prepared positive electrode active material includes lithium nickel-based composite oxide, and is in the form of single particles. The prepared positive electrode active material may exhibit a layered crystal structure, unlike the first fired product, which exhibits, for example, a hexagonal crystal structure and a cubic crystal structure. For example, the lithium nickel-based composite oxide may be represented by Chemical Formula 1.

$\text{Li}_{a1}\text{Ni}_{x1}\text{M}_{y1}\text{M}_{z1}\text{O}_{b1}\text{X}_{b1}$ Chemical Formula 1:

[0055] In Chemical Formula 1, $0.8 \leq a1 \leq 1.2$, $0.7 \leq x1 < 1$, $0 < y1 \leq 0.3$, $0 \leq z1 \leq 0.3$, $0.9 \leq x1 + y1 + z1 \leq 1.1$, and $0 \leq b1 \leq 0.1$, $\text{M}_{sup.1}$ and $\text{M}_{sup.2}$ each independently are or include at least one of Al, B, Ba, Ca, Ce, Co, Cr, Cu, Fe, Mg, Mn, Mo, Nb, Si, Sn, Sr, Ti, V, W, Y, Zn, and Zr, and X is or includes at least one of F, P, and S.

[0056] As an example, the lithium nickel-based composite oxide may be represented by Chemical Formula 2 or Chemical Formula 3.

$\text{Li}_{a2}\text{Ni}_{x2}\text{Co}_{y2}\text{M}_{z2}\text{O}_{b2}\text{X}_{b2}$ Chemical Formula 2:

[0057] In Chemical Formula 2, $0.9 \leq a2 \leq 1.2$, $0.7 \leq x2 < 1$, $0 < y2 \leq 0.3$, $0 \leq z2 \leq 0.3$, $0.9 \leq x2 + y2 + z2 \leq 1.1$, and $0 \leq b2 \leq 0.1$, $\text{M}_{sup.3}$ is or includes at least one of Al, B, Ba, Ca, Ce, Cr, Cu, Fe, Mg, Mn, Mo, Nb, Si, Sn, Sr, Ti, V, W, Y, Zn, and Zr, and X is or includes at least one of F, P, S, or a combination thereof.

[0058] In Chemical Formula 2 $0.8 \leq x2 \leq 0.99$, $0.01 \leq y2 \leq 0.2$, and $0 \leq z2 \leq 0.2$, or $0.9 \leq x2 \leq 0.99$, $0.01 \leq y2 \leq 0.1$, and $0 \leq z2 \leq 0.1$.

$\text{Li}_{a3}\text{Ni}_{x3}\text{CO}_{y3}\text{M}_{z3}\text{M}_{w3}\text{O}_{b3}\text{X}_{b3}$ Chemical Formula 3:

[0059] In Chemical Formula 3, $0.9 \leq a3 \leq 1.2$, $0.7 \leq x3 \leq 0.98$, $0.01 \leq y3 < 0.29$, $0.01 \leq z3 \leq 0.29$, $0 \leq w3 \leq 0.29$, $0.9 \leq x3 + y3 + z3 + w3 \leq 1.1$, and $0 \leq b3 \leq 0.1$, $\text{M}_{sup.4}$ is or includes at least one of Al, Mn, or a combination thereof, $\text{M}_{sup.5}$ is or includes at least one of B, Ba, Ca, Ce, Cr, Cu, Fe, Mg, Mo, Nb, Si, Sn, Sr, Ti, V, W, Y, Zn, and Zr, and X is or includes at least one of F, P, S, or a combination thereof.

[0060] In Chemical Formula 3 $0.8 \leq x3 \leq 0.98$, $0.01 \leq y3 \leq 0.19$, $0.01 \leq z3 \leq 0.19$, and $0 \leq w3 \leq 0.19$, $0.9 \leq x3 \leq 0.98$, $0.01 \leq y3 \leq 0.09$, $0.01 \leq z3 \leq 0.09$, and $0 \leq w3 \leq 0.09$.

[0061] Such a high nickel-based positive electrode active material, as described above, may require a washing process, because an excessive amount of residual lithium is generated on the particle surface during the synthesis process. The washing process may remove a lithium compound alone such as $\text{Li}_{2}\text{CO}_{3}$, LiOH, and the like, from the surface of positive electrode active material particles, but may also remove active lithium inside the positive electrode active material, which may change a crystal structure or cause a phase change as well as decrease reversible capacity, and thereby deteriorate performance. For example, lithium may be eluted and changed from lithium-nickel-oxides and the like to nickel-oxides and the like, for example, to form NiO, and accordingly, a type of reduction reaction may occur in which the oxidation number of Ni decreases from +3 to +2. The compounds such as NiO and the like may not be converted into lithium-containing compounds, and may thus promote a side reaction of the positive electrode active material particle surface with an electrolyte, as well as reduce reversible capacity, thereby, deteriorating overall performance. In order to overcome this challenge, after the washing process, a process of restoring the deteriorated surface through additional compensation of lithium may be performed, but another

challenge of causing an additional side reaction and increasing a processing cost may occur.

[0062] In examples, the novel method of preparing the positive electrode active material according to some example embodiments may omit the washing process, and may thus overcome the performance degradation challenge due to the washing, and may also decrease a residual lithium content on the surface to about 2000 ppm or less, preparing a positive electrode active material with desired or improved surface characteristics. The residual lithium content on the surface of the positive electrode active material may be, for example, less than or equal to about 2000 ppm, less than or equal to about 1900 ppm, or less than or equal to about 1800 ppm.

[0063] The average particle diameter (D_{50}) of single particles of the manufactured positive electrode active material may be about 1 μm to about 6 μm , or about 2 μm to about 5 μm . In addition, $(D_{90}-D_{10})/D_{50}$ of the single particles may be less than or equal to about 1.2. The average particle diameter (D_{50}) of the single particles indicates a size of particles having a cumulative volume of 50 volume % in the particle size distribution that is obtained by measuring the size of about 20 particles chosen at random from a SEM image. D_{90} is a particle size at a cumulative volume of about 90 volume % in the particle distribution, and D_{10} is a particle size at a cumulative volume of about 10 volume %.

[0064] In addition, the positive electrode active material may exhibit a low specific surface area and, for example, a BET specific surface area of the positive electrode active material may be about 0.5 to about 1.0 m^2/g , or about 0.6 to about 0.8 m^2/g .

[0065] The positive electrode active material may exhibit a low specific surface area and a clean surface, and may thus have a high pellet density. For example, the pellet density of the positive electrode active material may be about 2.7 to about 3.3 g/cc , or about 2.8 to about 3.4 g/cc . A rechargeable lithium battery, to which such a positive electrode active material is applied, may achieve a high energy density. The pellet density may be measured in the following method. The positive electrode active material is weighed by about 3 g, which is added to a mold (area: about 1.298 cm^2), and a mold bar is slowly inserted into a mold main body. The mold set is placed in a hydraulic press and pressurized at about 3 tons for about 30 seconds to measure a height and thus pellet density.

[0066] In addition, the positive electrode active material may have a tap density of about 1.5 g/cc to about 2.2 g/cc , for example about 1.7 g/cc to about 2.0 g/cc . The tap density may be, for example, measured by using a tap density measuring device (TAP-2S, Logan Instruments Co.), for example, filling about 50 cc of the positive electrode active material powder in a 100 cc-mass cylinder, about 1000 times, repeatedly tapping the positive electrode active material powder from a height of about 3 mm per second, and dividing the mass by a volume.

[0067] The positive electrode active material may have a ratio of peak intensity of a (003) plane to that of a (104) plane of $I_{(003)}/I_{(104)}$ in a range that is greater than about 1.25, for example greater than or equal to about 1.30, or greater than or equal to about 1.40.

[0068] On the other hand, in some example embodiments, the positive electrode active material obtained through the second heat treatment may be mixed with a coating raw material, and further thirdly heat-treated. Through this, desired coating may proceed on the surface of the positive electrode active material in a simple method. The coating raw material may include, for example, one or more of Al, B, Co, Mg, V, Zn, and Zr. Herein, a content of the coating elements is not particularly limited but may be, for example, about 0.1 wt % to about 5 wt % or about 0.5 wt % to about 3 wt % based on 100 wt % of the total metal excluding lithium in the final positive electrode active material. For example, the method for preparing a positive electrode active material according to some example embodiments may include mixing the positive electrode active material obtained through the second heat treatment with a boron raw material, and performing a third heat treatment within a temperature range of about 250° C. to about 450° C. for a period of about 2 hours to about 8 hours, through which a positive electrode active material with improved lifecycle characteristics may be prepared. Herein, a boron content may be about 0.01 wt % to about 0.5 wt %

or about 0.05 wt % to about 0.3 wt % based on 100 wt % of the total metal excluding lithium in the final positive electrode active material.

Positive Electrode

[0069] In some example embodiments, a positive electrode for a rechargeable lithium battery including the aforementioned positive electrode active material is provided. For example, the positive electrode includes a positive electrode current collector and a positive electrode active material layer on the positive electrode current collector, and the positive electrode active material layer may include the aforementioned positive electrode active material. The positive electrode active material layer may further include other types of positive electrode active materials in addition to the aforementioned positive electrode active material, and may optionally further include a binder, a conductive material, or a combination thereof.

Binder

[0070] The binder improves binding properties of positive electrode active material particles with one another and with a current collector. Examples of the binder may include at least one of polyvinyl alcohol, carboxymethyl cellulose, hydroxypropyl cellulose, diacetyl cellulose, polyvinyl chloride, carboxylated polyvinyl chloride, polyvinyl fluoride, an ethylene oxide-containing polymer, polyvinylpyrrolidone, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, a styrene-butadiene rubber, a (meth)acrylated styrene-butadiene rubber, an epoxy resin, a (meth)acrylic resin, a polyester resin, and nylon, but are not limited thereto.

Conductive Material

[0071] The conductive material is included to provide electrode conductivity, and any electrically conductive material may be a conductive material unless the electrically conductive material causes a chemical change in the battery. Examples of the conductive material may include a carbon-based material such as at least one of natural graphite, artificial graphite, carbon black, acetylene black, ketjen black, a carbon fiber, a carbon nanofiber, a carbon nanotube, and the like; a metal-based material of a metal powder or a metal fiber including at least one of copper, nickel, aluminum, silver, and the like; a conductive polymer such as a polyphenylene derivative; or a mixture thereof.

[0072] Each content of the binder and the conductive material may be about 0.5 wt % to about 5 wt % based on 100 wt % of the positive electrode active material layer.

[0073] The positive electrode current collector may include Al, but is not limited thereto.

[0074] The positive electrode according to some example embodiments may achieve high mixture density. For example, a density of the positive electrode active material layer may be about 3.35 to about 3.85 g/cc or about 3.45 to about 3.75 g/cc. The density of the positive electrode active material layer refers to the ratio of the weight to the volume of the pressed positive active material layer. The density of the positive electrode active material layer can be measured by measuring a cross-sectional area, thickness and weight of the positive active material layer excluding the positive electrode current collector, calculating the volume by multiplying the cross-sectional area and the thickness, and dividing the weight by the volume. The positive electrode active material according to some example embodiments may be advantageous to achieve the positive electrode densities, and a positive electrode, which satisfy the positive electrode density ranges, and may be appropriate for achieving a high-capacity and high-energy density rechargeable lithium battery.

Rechargeable Lithium Battery

[0075] Some example embodiments include a rechargeable lithium battery including the aforementioned positive electrode, negative electrode, and electrolyte. As an example, the rechargeable lithium battery may include a positive electrode, a negative electrode, a separator between the positive electrode and the negative electrode, and an electrolyte solution.

[0076] The rechargeable lithium battery may be classified into cylindrical, prismatic, pouch, coin, etc. depending on the shape. FIGS. 1 to 4 are schematic views illustrating the rechargeable lithium battery according to some example embodiments, where FIG. 1 illustrates a cylindrical battery,

FIG. 2 illustrates a prismatic battery, and FIGS. 3 and 4 illustrate a pouch-shaped battery. Referring to FIGS. 1 to 4, the rechargeable lithium battery 100 includes an electrode assembly 40 with a separator 30 interposed between the positive electrode 10 and the negative electrode 20, and a case 50 in which the electrode assembly 40 is housed. The positive electrode 10, the negative electrode 20, and the separator 30 may be impregnated with an electrolyte solution (not shown). The rechargeable lithium battery 100 may include a sealing member 60 that seals the case 50 as shown in FIG. 1. Additionally, in FIG. 2, the rechargeable lithium battery 100 may include a positive electrode lead tab 11, a positive electrode terminal 12, a negative lead tab 21, and a negative electrode terminal 22. As shown in FIGS. 3 and 4, the rechargeable lithium battery 100 includes an electrode tab 70 illustrated in FIG. 4, or a positive electrode tab 71 and a negative electrode tab 72 illustrated in FIG. 3, the electrode tabs 70/71/72 forming an electrical path for inducing the current formed in the electrode assembly 40 to the outside of the battery 100.

Negative Electrode

[0077] The negative electrode may include a current collector and a negative electrode active material layer on the current collector, and the negative electrode active material layer may further include a negative electrode active material, a binder, a conductive material, or a combination thereof.

Negative Electrode Active Material

[0078] The negative electrode active material may include at least one of a material that reversibly intercalates/deintercalates lithium ions, a lithium metal, a lithium metal alloy, a material capable of doping/dedoping lithium, or transition metal oxide.

[0079] The material that reversibly intercalates/deintercalates lithium ions may include, for example crystalline carbon, amorphous carbon, or a combination thereof as a carbon-based negative electrode active material. The crystalline carbon may be irregular, or sheet, flake, spherical, or fiber shaped natural graphite or artificial graphite. The amorphous carbon may be or include at least one of a soft carbon, a hard carbon, a mesophase pitch carbonization product, calcined coke, and the like.

[0080] The lithium metal alloy includes an alloy of lithium and a metal including at least one of Na, K, Rb, Cs, Fr, Be, Mg, Ca, Sr, Si, Sb, Pb, In, Zn, Ba, Ra, Ge, Al, and Sn.

[0081] The material capable of doping/dedoping lithium may be or include a Si-based negative electrode active material or a Sn-based negative electrode active material. The Si-based negative electrode active material may include at least one of silicon, a silicon-carbon composite, $\text{SiO}_{0.5}$ ($0 < x \leq 2$), a Si-Q alloy (wherein Q is or includes at least one of an alkali metal, an alkaline-earth metal, a Group 13 element, a Group 14 element (excluding Si), a Group 15 element, a Group 16 element, a transition metal, a rare earth element, and a combination thereof, for example at least one of Mg, Ca, Sr, Ba, Ra, Sc, Y, Ti, Zr, Hf, Rf, V, Nb, Ta, Db, Cr, Mo, W, Sg, Tc, Re, Bh, Fe, Pb, Ru, Os, Hs, Rh, Ir, Pd, Pt, Cu, Ag, Au, Zn, Cd, B, Al, Ga, Sn, In, Tl, Ge, P, As, Sb, Bi, S, Se, Te, Po, and a combination thereof), or a combination thereof. The Sn-based negative electrode active material may be or include at least one of Sn, $\text{SnO}_{0.5}$, a Sn alloy, or a combination thereof.

[0082] The silicon-carbon composite may be or include a composite of silicon and amorphous carbon. An average particle diameter (D_{50}) of the silicon-carbon composite particles may be, for example, about 0.5 μm to about 20 μm . According to some example embodiments, the silicon-carbon composite may be in the form of silicon particles and amorphous carbon coated on the surface of the silicon particles. For example, the silicon-carbon composite may include a secondary particle (core) in which silicon primary particles are assembled and an amorphous carbon coating layer (shell) on the surface of the secondary particle. The amorphous carbon may also be present between the silicon primary particles, for example, the silicon primary particles may be coated with amorphous carbon. The secondary particles may be dispersed in an amorphous carbon matrix.

[0083] The silicon-carbon composite may further include crystalline carbon. For example, the silicon-carbon composite may include a core including crystalline carbon and silicon particles and

an amorphous carbon coating layer on the surface of the core. The crystalline carbon may be or include artificial graphite, natural graphite, or a combination thereof. The amorphous carbon may include at least one of soft carbon or hard carbon, a mesophase pitch carbonized product, and calcined coke.

[0084] When the silicon-carbon composite includes silicon and amorphous carbon, a content of silicon may be about 10 wt % to about 50 wt % and a content of amorphous carbon may be about 50 wt % to about 90 wt % based on 100 wt % of the silicon-carbon composite. In addition, when the composite includes silicon, amorphous carbon, and crystalline carbon, a content of silicon may be about 10 wt % to about 50 wt %, a content of crystalline carbon may be about 10 wt % to about 70 wt %, and a content of amorphous carbon may be about 20 wt % to about 40 wt % based on 100 wt % of the silicon-carbon composite.

[0085] Additionally, a thickness of the amorphous carbon coating layer may be about 5 nm to about 100 nm. An average particle diameter ($D_{.50}$) of the silicon particles (primary particles) may be about 10 nm to about 1 μm , or about 10 nm to about 200 nm. The silicon particles may be present as silicon alone, in the form of a silicon alloy, or in an oxidized form. The oxidized form of silicon may be represented by SiO_x ($0 < x \leq 2$). For example, the atomic content ratio of Si: O, which indicates a degree of oxidation, may be about 99:1 to about 33:67. As used herein, when a definition is not otherwise provided, an average particle diameter ($D_{.50}$) indicates a particle where an accumulated volume is about 50 volume % in a particle distribution.

[0086] The Si-based negative electrode active material, or Sn-based negative electrode active material, may be mixed with the carbon-based negative electrode active material. When the Si-based negative electrode active material or Sn-based negative electrode active material and the carbon-based negative electrode active material are mixed, the mixing ratio may be a weight ratio of about 1:99 to about 90:10.

Binder

[0087] The binder is configured to adhere the negative electrode active material particles to each other, and also to adhere the negative electrode active material to the current collector. The binder may be or include a non-aqueous binder, an aqueous binder, a dry binder, or a combination thereof.

[0088] The non-aqueous binder may include at least one of polyvinyl chloride, carboxylated polyvinyl chloride, polyvinyl fluoride, an ethylene propylene copolymer, polystyrene, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, polyamideimide, polyimide, or a combination thereof.

[0089] The aqueous binder may include at least one of a styrene-butadiene rubber, a (meth)acrylated styrene-butadiene rubber, a (meth)acrylonitrile-butadiene rubber, a (meth)acrylic rubber, a butyl rubber, a fluorine rubber, polyethylene oxide, polyvinylpyrrolidone, polyepichlorohydrin, polyphosphazene, poly(meth)acrylonitrile, an ethylene propylene diene copolymer, polyvinylpyridine, chlorosulfonated polyethylene, latex, a polyester resin, a (meth)acrylic resin, a phenol resin, an epoxy resin, polyvinyl alcohol, or a combination thereof.

[0090] When an aqueous binder is used as the negative electrode binder, a cellulose-based compound configured to impart viscosity may be further included. As the cellulose-based compound, one or more of carboxymethyl cellulose, hydroxypropylmethyl cellulose, methyl cellulose, or alkali metal salts thereof may be mixed and used. The alkali metal may be or include at least one of Na, K, or Li.

[0091] The dry binder may be or include a polymer material capable of being in the form of fiber, and may be or include, for example, at least one of polytetrafluoroethylene, polyvinylidene fluoride, a polyvinylidene fluoride-hexafluoropropylene copolymer, polyethylene oxide, or a combination thereof.

Conductive Material

[0092] The conductive material is included to provide electrode conductivity, and any electrically conductive material may be used as a conductive material unless the electrically conductive

material causes a chemical change. Examples of the conductive material include a carbon-based material such as at least one of natural graphite, artificial graphite, carbon black, acetylene black, ketjen black, a carbon fiber, a carbon nanofiber, a carbon nanotube, and the like; a metal-based material of a metal powder or a metal fiber including at least one of copper, nickel, aluminum silver, and the like; a conductive polymer such as a polyphenylene derivative; or a mixture thereof. [0093] A content of the negative electrode active material may be about 95 wt % to about 99.5 wt % based on 100 wt % of the negative electrode active material layer, and a content of the binder may be about 0.5 wt % to about 5 wt % based on 100 wt % of the negative electrode active material layer. For example, the negative electrode active material layer may include about 90 wt % to about 99 wt % of the negative electrode active material, about 0.5 wt % to about 5 wt % of the binder, and about 0.5 wt % to about 5 wt % of the conductive material.

Current Collector

[0094] The negative electrode current collector may include, for example, at least one of indium (In), copper (Cu), magnesium (Mg), stainless steel, titanium (Ti), iron (Fe), cobalt (Co), nickel (Ni), zinc (Zn), aluminum (Al), germanium (Ge), lithium (Li), or an alloy thereof, and may be in the form of a foil, sheet, or foam. A thickness of the negative electrode current collector may be, for example, about 1 μm to about 20 μm , about 5 μm to about 15 μm , or about 7 μm to about 10 μm .

Electrolyte

[0095] For example, the electrolyte for a rechargeable lithium battery may be an electrolyte solution, which may include a non-aqueous organic solvent and a lithium salt.

[0096] The non-aqueous organic solvent may constitute a medium for transmitting ions taking part in the electrochemical reaction of a battery. The non-aqueous organic solvent may be or include at least one of a carbonate-based, ester-based, ether-based, ketone-based, or alcohol-based solvent, an aprotic solvent, or a combination thereof.

[0097] The carbonate-based solvent may include at least one of dimethyl carbonate (DMC), diethyl carbonate (DEC), dipropyl carbonate (DPC), methylpropyl carbonate (MPC), ethylpropyl carbonate (EPC), methylethyl carbonate (MEC), ethylene carbonate (EC), propylene carbonate (PC), butylene carbonate (BC), and the like. The ester-based solvent may include at least one of methyl acetate, ethyl acetate, n-propyl acetate, dimethyl acetate, methyl propionate, ethyl propionate, decanolide, mevalonolactone, valerolactone, caprolactone, and the like. The ether-based solvent may include at least one of dibutyl ether, tetraglyme, diglyme, dimethoxyethane, 2-methyltetrahydrofuran, 2,5-dimethyltetrahydrofuran, tetrahydrofuran, and the like. In addition, the ketone-based solvent may include cyclohexanone, and the like. The alcohol-based solvent may include at least one of ethanol, isopropyl alcohol, and the like and the aprotic solvent may include at least one of nitriles such as $\text{R}-\text{CN}$ (wherein R is a C2 to C20 linear, branched, or cyclic hydrocarbon group, a double bond, an aromatic ring, or an ether group, and the like; amides such as dimethylformamide; dioxolanes such as 1,3-dioxolane, 1,4-dioxolane, and the like; sulfolanes, and the like.

[0098] The non-aqueous organic solvent can be used alone or in a mixture of two or more types of solvents, and when two or more types of solvents are used in a mixture, a mixing ratio can be appropriately adjusted according to the desired battery performance, which may be known to those working in the field.

[0099] When using a carbonate-based solvent, a cyclic carbonate and a chain carbonate may be mixed, and the cyclic carbonate and the chain carbonate may be mixed in a volume ratio of about 1:1 to about 1:9.

[0100] The non-aqueous organic solvent may further include an aromatic hydrocarbon-based organic solvent. For example, a carbonate-based solvent and an aromatic hydrocarbon-based organic solvent may be mixed and used in a volume ratio of about 1:1 to about 30:1.

[0101] The electrolyte solution may further include at least one of vinyl ethylene carbonate, vinylene carbonate, or an ethylene carbonate-based compound to improve battery lifecycle.

[0102] Examples of the ethylene carbonate-based compound may include at least one of fluoroethylene carbonate, difluoroethylene carbonate, chloroethylene carbonate, dichloroethylene carbonate, bromoethylene carbonate, dibromoethylene carbonate, nitroethylene carbonate, and cyanoethylene carbonate.

[0103] The lithium salt dissolved in the organic solvent supplies lithium ions in a battery, enables a basic operation of a rechargeable lithium battery, and improves transportation of the lithium ions between positive and negative electrodes. Examples of the lithium salt may include at least one of LiPF_6 , LiBF_4 , LiSbF_6 , LiAsF_6 , LiClO_4 , LiAlO_2 , LiAlCl_4 , LiPO_2F_2 , LiCl , LiI , $\text{LiN}(\text{SO}_3\text{C}_2\text{F}_5)_2$, $\text{Li}(\text{FSO}_2)_2\text{N}$ (lithium bis(fluorosulfonyl) imide; LiFSI), $\text{LiC}_4\text{F}_9\text{SO}_3$, $\text{LiN}(\text{C}_x\text{F}_{2x+1}\text{SO}_2)_2$ ($\text{C}_y\text{F}_{2y+1}\text{SO}_2$) (wherein x and y are integers of 1 to 20), lithium trifluoromethane sulfonate, lithium tetrafluoroethane sulfonate, lithium difluorobis(oxalato) phosphate (LiDFBOP), and lithium bis(oxalato) borate (LiBOB).

[0104] A concentration of lithium salt may be within the range of about 0.1 M to about 2.0 M. When the concentration of lithium salt is within the above range, the electrolyte solution has a desired or appropriate ionic conductivity and viscosity, and thus desired or improved performance can be achieved and lithium ions can move effectively.

Separator

[0105] Depending on the type of the rechargeable lithium battery, a separator may be present between the positive electrode and the negative electrode. The separator may include at least one of polyethylene, polypropylene, polyvinylidene fluoride, or a multilayer film of two or more layers thereof, and a mixed multilayer film such as a polyethylene/polypropylene two-layer separator, polyethylene/polypropylene/polyethylene three-layer separator, polypropylene/polyethylene/polypropylene three-layer separator, and the like.

[0106] The separator may include a porous substrate and a coating layer including an organic material, an inorganic material, or a combination thereof on one or both surfaces of the porous substrate.

[0107] The porous substrate may be or include a polymer film formed of or including any one polymer including at least one of polyolefin such as polyethylene and polypropylene, polyester such as polyethylene terephthalate and polybutylene terephthalate, polyacetal, polyamide, polyimide, polycarbonate, polyether ketone, polyarylether ketone, polyether ketone, polyetherimide, polyamideimide, polybenzimidazole, polyethersulfone, polyphenylene oxide, a cyclic olefin copolymer, polyphenylene sulfide, polyethylene naphthalate, a glass fiber, TEFLON, and polytetrafluoroethylene, or a copolymer or mixture of two or more thereof.

[0108] The porous substrate may have a thickness of about 1 μm to about 40 μm , for example, about 1 μm to about 30 μm , about 1 μm to about 20 μm , about 5 μm to about 15 μm , or about 10 μm to about 15 μm .

[0109] The organic material may include a (meth)acrylic copolymer including a first structural unit derived from (meth)acrylamide, and a second structural unit including at least one of a structural unit derived from (meth)acrylic acid or (meth)acrylate, and a structural unit derived from (meth)acrylamidosulfonic acid or a salt thereof.

[0110] The inorganic material may include inorganic particles such as at least one of Al_2O_3 , SiO_2 , TiO_2 , SnO_2 , CeO_2 , MgO , NiO , CaO , GaO , ZnO , ZrO_2 , Y_2O_3 , SrTiO_3 , BaTiO_3 , $\text{Mg}(\text{OH})_2$, boehmite, and a combination thereof, but is not limited thereto. An average particle diameter (D_{50}) of the inorganic particles may be about 1 nm to about 2000 nm, for example, about 100 nm to about 1000 nm, or about 100 nm to about 700 nm.

[0111] The organic material and the inorganic material may be mixed in one coating layer, or a coating layer including an organic material and a coating layer including an inorganic material may be stacked.

[0112] The thickness of the coating layer may be about 0.5 μm to about 20 μm , for example, about 1 μm to about 10 μm , or about 1 μm to about 5 μm .

[0113] Examples and comparative examples of the present disclosure are described below.

However, the following examples are only examples of the present disclosure, and the present disclosure is not limited to the following examples.

EXAMPLES

Preparation of Positive Electrode Active Material

[0114] Nickel-based hydroxide ($\text{Ni}_{0.94}\text{Co}_{0.04}\text{Mn}_{0.02}(\text{OH})_2$) with an average particle diameter (D_{50}) of about 3 μm , anhydrous lithium hydroxide as a first lithium raw material, and ZrO_2 were mixed and then, subjected to a first heat treatment at 750° C. for 12 hours. Herein, the first lithium raw material was mixed so that a molar ratio (L_1) of lithium to a total metal of the nickel-based hydroxide was 0.6. In addition, ZrO_2 was mixed so that Zr was included in an amount of 0.125 wt % relative to 100 wt % of a sum of total metal of the nickel-based hydroxide and Zr of ZrO_2 .

[0115] An SEM image of a first fired product obtained through the first heat treatment is shown in FIG. 5, and its XRD Rietveld Refinement result is shown in FIG. 6. Referring to FIG. 5, the first fired product was in a form of secondary particles formed by agglomerating a plurality of primary particles and having a size of about 2 μm to 3 μm . Referring to FIG. 6, the first fired product was confirmed to have 89.3% of a hexagonal crystal structure and 10.7% of a cubic crystal structure.

[0116] The first fired product, anhydrous lithium hydroxide as a second lithium raw material, and cobalt oxide as a coating raw material were mixed, and secondarily heat-treated at 780° C. for 16 hours. Herein, a molar ratio (L_2) of lithium in the second lithium raw material to the total metal excluding lithium was set to be 0.45 in the first fired product, and also, in the first fired product, a cobalt content was designed to be 2.0 wt % based on 100 wt % of the total metal excluding lithium in the positive electrode active material.

[0117] After the second heat treatment, pulverization was performed using an air flow pulverization method to obtain a positive electrode active material in the form of single particles.

[0118] The SEM images of the final positive electrode active material in the form of single particles are shown in FIGS. 7 and 8. Referring to FIGS. 7 and 8, the positive electrode active material was confirmed to be in the form of single particles with a substantially uniform particle size and a clean surface. In addition, as the XRD Rietveld Refinement result of the final positive electrode active material, the positive electrode active material was confirmed to have a hexagonal crystal structure of 100%.

Residual Lithium and XRD Analysis

[0119] The first fired product and the final positive electrode active material of the example were analyzed with respect to a content of residual lithium through a pH titration method, and an XRD analysis was performed to obtain a full width at half maximum (FWHM) on a (003) plane and a (104) plane and a ratio of peak intensity of the (003) plane to that of the (104) plane, and the results are shown in Table 1.

TABLE-US-00001
TABLE 1 Final positive electrode First fired product active material LiOH (wt %)
0.190 0.388 Li_2CO_3 (wt %) 0.312 0.308 Residual Li (ppm) 1135 1704 FWHM.sub.(003) — 0.127 FWHM.sub.(104) — 0.139 I.sub.(003)/I.sub.(104) 0.493 1.433

[0120] Referring to Table 1, in the final positive electrode active material, a content of the residual lithium was confirmed to be about 1704 ppm. The Example, even though the high nickel-based positive electrode active material was synthesized without the washing process, exhibited a low residual lithium content.

[0121] In addition, referring to Table 1, the final positive electrode active material exhibited a $I_{(003)}/I_{(104)}$ ratio of 1.433 in the XRD analysis.

[0122] While this disclosure has been described in connection with what is presently considered to be practical example embodiments, it is to be understood that the disclosure is not limited to the

disclosed example embodiments. On the contrary, the disclosure is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims.

DESCRIPTION OF SYMBOLS

[0123] **100**: rechargeable lithium battery **10**: positive electrode [0124] **11**: positive electrode lead tab **12**: positive electrode terminal [0125] **20**: negative electrode **21**: negative electrode lead tab [0126] **22**: negative electrode terminal **30**: separator [0127] **40**: electrode assembly **50**: case [0128] **60**: sealing member **70**: electrode tab [0129] **71**: positive electrode tab **72**: negative electrode tab

Claims

1. A method of preparing a positive electrode active material, the method comprising: mixing a nickel-based precursor and a first lithium raw material, and performing a first heat treatment on the mixed nickel-based precursor and first lithium raw material at a temperature in a range of about 500° C. to about 750° C. to prepare a first fired product, and mixing the first fired product and a second lithium raw material, and performing a second heat treatment on the mixed first fired product and second lithium raw material at a temperature in a range of about 650° C. to about 850° C. to prepare a positive electrode active material that comprises single particles including a lithium nickel-based composite oxide, wherein: a molar ratio ($L_{sup.1}$) of lithium in the first lithium raw material to a total metal of the nickel-based precursor is about 0.2 to about 0.9, a molar ratio ($L_{sup.2}$) of lithium in the second lithium raw material to a total metal excluding lithium in the first fired product is about 0.1 to about 0.8, and about $0.9 \leq L_{sup.1} + L_{sup.2} \leq \text{about } 1.1$.
2. The method as claimed in claim 1, wherein $L_{sup.1}$ is about 0.4 to about 0.7, and $L_{sup.2}$ is about 0.3 to about 0.6.
3. The method as claimed in claim 1, wherein $L_{sup.1} > L_{sup.2}$.
4. The method as claimed in claim 1, wherein $L_{sup.1} \leq L_{sup.2}$.
5. The method as claimed in claim 1, wherein the nickel-based precursor is represented by Chemical Formula 11 or Chemical Formula 12:
$$Ni_{sub.x11}M_{sup.11.sub.y11}M_{sup.12.sub.z11}(OH)_{sub.2}$$
 Chemical Formula 11 wherein, in Chemical Formula 11, $0.7 \leq x_{11} < 1$, $0 < y_{11} \leq 0.3$, $0 \leq z_{11} \leq 0.3$, $0.9 \leq x_{11} + y_{11} + z_{11} \leq 1.1$, and $M_{sup.11}$ and $M_{sup.12}$ each independently comprise one or more of Al, B, Ba, Ca, Ce, Co, Cr, Cu, Fe, Mg, Mn, Mo, Nb, Si, Sn, Sr, Ti, V, W, Y, Zn, and Zr,
$$Ni_{sub.x12}M_{sup.13.sub.y12}M_{sup.14.sub.z12}O_{sub.2-b12}X_{sub.b12}$$
 Chemical Formula 12 wherein, in Chemical Formula 12, $0.7 \leq x_{12} < 1$, $0 < y_{12} \leq 0.3$, $0 \leq z_{12} \leq 0.3$, $0.9 \leq x_{12} + y_{12} + z_{12} \leq 1.1$, $0 \leq b_{12} \leq 0.1$, $M_{sup.13}$ and $M_{sup.14}$ each independently comprise one or more of Al, B, Ba, Ca, Ce, Co, Cr, Cu, Fe, Mg, Mn, Mo, Nb, Si, Sn, Sr, Ti, V, W, Y, Zn, and Zr, and X comprises at least one of F, P, and S.
6. The method as claimed in claim 1, wherein: the nickel-based precursor comprises secondary particles formed by agglomerating a plurality of primary particles, and an average particle diameter ($D_{sub.50}$) of the secondary particles is about 1 μm to about 6 μm , and a difference between the average particle diameter ($D_{sub.50}$) of the secondary particles in the nickel-based precursor and an average particle diameter ($D_{sub.50}$) of the single particles in the positive electrode active material is less than or equal to about 1 μm .
7. The method as claimed in claim 1, wherein the first lithium raw material and the second lithium raw material comprise anhydrous lithium hydroxide.
8. The method as claimed in claim 1, wherein: the first heat treatment is performed at a temperature in a range of about 550° C. to about 750° C. for a period of about 6 hours to 12 hours in an oxidizing atmosphere, and the first heat treatment temperature is lower than the second heat treatment temperature.
9. The method as claimed in claim 1, wherein: mixing the nickel-based precursor and the first

lithium raw material further comprises mixing zirconium raw material, and a zirconium content of the zirconium raw material is about 0.01 wt % to about 1 wt % based on 100 wt % of a total metal content of the nickel-based precursor and the zirconium of the zirconium raw material.

10. The method as claimed in claim 1, wherein: the first fired product comprises secondary particles formed by agglomerating a plurality of primary particles, and an average particle diameter ($D_{sub.50}$) of the secondary particles is about 1 μm to about 6 μm .

11. The method as claimed in claim 1, wherein: the first fired product comprises a lithium nickel-based composite oxide, and the lithium nickel-based composite oxide comprises about 60 vol % to about 95 vol % of a hexagonal crystal structure and about 5 vol % to about 40 vol % of a cubic crystal structure.

12. The method as claimed in claim 1, wherein the second heat treatment is performed at a temperature in a range of about 750° C. to about 810° C. for a period of about 10 hours to about 16 hours in an oxidizing atmosphere.

13. The method as claimed in claim 1, wherein: mixing the first fired product and the second lithium raw material further comprises mixing a coating raw material, and the coating raw material comprises one or more of Al, B, Co, Mg, V, Zn, and Zr.

14. The method as claimed in claim 1, wherein the method does not include washing the positive electrode active material with water.

15. The method as claimed in claim 1, wherein the prepared positive electrode active material is represented by Chemical Formula 1:

$\text{Li}_{sub.a1}\text{Ni}_{sub.x1}\text{M}_{sup.1sub.y1}\text{M}_{sup.2sub.z1}\text{O}_{sub.2-b1}\text{X}_{sub.b1}$ Chemical Formula 1:
wherein, in Chemical Formula 1, $0.8 \leq a1 \leq 1.2$, $0.7 \leq x1 < 1$, $0 < y1 \leq 0.3$, $0 \leq z1 \leq 0.3$, $0.9 \leq x1 + y1 + z1 \leq 1.1$, and $0 \leq b1 \leq 0.1$, $\text{M}_{sup.1}$ and $\text{M}_{sup.2}$ each independently comprise at least one of Al, B, Ba, Ca, Ce, Co, Cr, Cu, Fe, Mg, Mn, Mo, Nb, Si, Sn, Sr, Ti, V, W, Y, Zn, and Zr, and X comprises at least one of F, P, and S.

16. The method as claimed in claim 1, wherein an average particle diameter ($D_{sub.50}$) of the single particles of the prepared positive electrode active material is about 1 μm to about 6 μm , a BET specific surface area is about 0.5 to about 1.0 $\text{m}_{sup.2}/\text{g}$, and a residual lithium content on surface thereof is less than or equal to about 2000 ppm.

17. The method as claimed in claim 1, wherein a pellet density of the prepared positive electrode active material is about 2.7 g/cc to about 3.3 g/cc, and the tap density of the prepared positive electrode active material is about 1.5 g/cc to about 2.2 g/cc.

18. The method as claimed in claim 1, wherein the prepared positive electrode active material has a ratio of the peak intensity of a (003) plane to a peak intensity of a (104) plane that is greater than or equal to about 1.25 in X-ray diffraction analysis.

19. The method as claimed in claim 1, further comprising mixing the prepared positive electrode active material and a coating raw material, and subjecting the prepared positive electrode active material and the coating raw material to a third heat treatment, the coating raw material comprising one or more of Al, B, Co, Mg, V, Zn, and Zr.

20. A rechargeable lithium battery, comprising a positive electrode including the positive electrode active material prepared according to the method as claimed in claim 1, a negative electrode, and an electrolyte.
